

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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Date	
Time	

COURSE NAME: Cybersecurity Incident Handling And Response

COURSE CODE: CSA60

COURSE OUTCOME COVERED

CO2: *Analyze incident response preparation strategies through examination of team roles, policies, SOPs, risk management practices, along with compliance frameworks to ensure organizational readiness. (BL3, BL6)*

Activity Tool : Incident Response Team Planning and SOP Development (High)

Assessment Tool Weightage: 40%

Code of Conduct

I _____ Shaik reehan _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Shaik reehan

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Incident Response Team Planning	20%	Clearly identifies all key Incident Response Team roles, responsibilities, reporting structure, and interdepartmental coordination.	Identifies most roles and responsibilities with minor omissions.	Identifies only some roles with limited understanding of responsibilities.	Unable to identify appropriate team roles or responsibilities.
SOP Development and Incident Handling Process	25%	Develops a comprehensive, logical SOP covering detection, reporting, containment, eradication, recovery, documentation, and communication.	SOP includes most required phases with minor omissions or sequencing errors.	SOP covers only basic steps with several missing components.	SOP is incomplete, incorrect, or lacks logical flow.
Security Policies, Awareness, and Governance	20%	Presents well-structured security policies, awareness program, and governance practices with clear justification.	Includes relevant policies and awareness activities with minor gaps.	Provides limited policies or awareness measures with insufficient explanation.	Fails to identify appropriate policies or governance practices.
Risk Assessment, BCP, and DRP	20%	Accurately identifies critical assets, performs risk assessment, and clearly explains BCP and DRP strategies with appropriate recommendations.	Performs basic risk assessment and explains BCP/DRP with minor omissions.	Partial risk assessment with limited understanding of BCP and DRP concepts.	Unable to perform risk assessment or explain BCP/DRP.
Documentation, Communication, and Presentation	15%	Documentation is well-organized, professionally presented, uses diagrams/tables effectively, and	Documentation is organized with minor formatting or clarity issues.	Documentation is partially organized with several	Documentation is poorly organized, incomplete, or

		communicates ideas clearly.		presentation issues.	difficult to understand.
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5. Standard Operating Procedures (SOPs)

Introduction

A Standard Operating Procedure (SOP) is a documented set of step-by-step instructions that guides the Incident Response Team (IRT) in responding to cybersecurity incidents. During a ransomware attack, an SOP helps ensure that every team member understands their responsibilities, reduces confusion, minimizes downtime, protects business data, and restores critical systems safely. A manufacturing company depends on continuous operations, so a quick and organized response is essential.

1. Incident Detection

The first step is identifying the ransomware attack. Detection may occur through security monitoring tools, antivirus software, SIEM alerts, unusual system behavior, encrypted files, or user reports. Once detected, the Incident Response Team should immediately verify the attack and determine its scope.

Responsibilities

- Security Analysts monitor alerts and verify the ransomware.
 - Employees immediately report suspicious activities.
 - Incident Response Manager confirms the incident and activates the response process.
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2. Incident Reporting and Initial Assessment

After confirmation, the incident should be formally reported according to the organization's incident response plan. The Incident Response Manager assesses the severity by identifying affected servers, business impact, data loss, and whether the attack is spreading.

Responsibilities

- Incident Response Manager classifies the incident.
 - Security Analysts gather preliminary information.
 - Management receives an initial situation report.
 - Documentation begins for future investigation.
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3. Containment

The primary goal of containment is to stop the ransomware from spreading to other servers, workstations, and network devices. Infected systems should be isolated from the network while ensuring unaffected systems continue operating whenever possible.

Responsibilities

- Network Administrators disconnect infected systems.
 - Disable compromised user accounts.
 - Block malicious IP addresses and domains.
 - Stop unnecessary network services if required.
 - Preserve evidence without modifying infected systems.
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4. Investigation and Analysis

The Incident Response Team investigates how the ransomware entered the network. They analyze security logs, identify the attack vector, determine which systems are affected, and collect digital evidence for future analysis and legal purposes.

Responsibilities

- Digital Forensics Team collects and preserves evidence.
 - Security Analysts examine logs and Indicators of Compromise (IOCs).
 - Threat Intelligence Team identifies ransomware variants.
 - Legal Team ensures evidence handling follows legal requirements.
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5. Eradication

Once the investigation is complete, the ransomware and all malicious files must be completely removed. Vulnerabilities exploited by attackers should also be fixed to prevent reinfection.

Responsibilities

- Remove ransomware using approved security tools.
 - Delete malicious files and processes.
 - Install operating system and application patches.
 - Reset compromised passwords.
 - Update antivirus and endpoint protection software.
 - Verify systems are malware-free.
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6. Recovery

The organization restores affected systems using verified backups and carefully returns them to normal operation. Before reconnecting systems to the production network, they should be tested to ensure they are secure.

Responsibilities

- Restore data from clean backups.
 - Test applications and production servers.
 - Monitor network traffic for suspicious activity.
 - Confirm business services operate normally.
 - Obtain management approval before full production resumes.
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7. Post-Incident Review

After recovery, the organization conducts a lessons-learned meeting to improve future incident response. The team evaluates what worked well, identifies weaknesses, and updates security controls and procedures.

Responsibilities

- Prepare the final incident report.
 - Update the Incident Response Plan.
 - Improve security monitoring.
 - Conduct employee awareness training.
 - Perform regular backup and recovery testing.
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Conclusion

A well-defined Standard Operating Procedure enables the Incident Response Team to respond quickly and effectively to ransomware attacks. By following the stages of detection, reporting, containment, investigation, eradication, recovery, and post-incident review, the organization minimizes financial loss, reduces downtime, protects sensitive information, and strengthens its cybersecurity posture.

6. Escalation Matrix and Communication Channels

Introduction

An Escalation Matrix defines how cybersecurity incidents are reported and who should be notified at each severity level. In a university, where student records contain sensitive personal information, timely communication between technical teams, management, and legal authorities is essential. A clear reporting hierarchy prevents delays, avoids confusion, and ensures a coordinated response.

1. Escalation Matrix

Severity	Incident Type	Personnel to Inform	Response Time
Level 1 (Low)	Single malware infection	IT Help Desk, System Administrator	Within 30 minutes
Level 2 (Medium)	Multiple systems affected	Security Team, IT Manager	Within 15 minutes
Level 3 (High)	Student database compromised	Incident Response Manager, Legal Team, HR	Immediately
Level 4 (Critical)	Major data breach or ransomware	Executive Management, CISO, Regulatory Authorities	Immediate emergency response

2. Communication Channels

The organization should use secure and reliable communication methods during an incident.

- Official Email for status updates.
- Secure Messaging Applications (Microsoft Teams, Slack, etc.).
- Incident Ticketing System for tracking response activities.
- Emergency Phone Calls for critical incidents.
- Video Conferencing for Incident Response Team meetings.
- Management Dashboards for real-time monitoring.

Sensitive information should only be shared through encrypted and authorized communication channels.

3. Reporting Hierarchy

Employee/User



IT Help Desk



System Administrator



Security Analysts



Incident Response Manager



Chief Information Security Officer (CISO) / IT Director



University Executive Management



Legal Authorities / Regulatory Agencies (if required)

Each level should receive accurate information without unnecessary delays.

4. Responsibilities

Technical Team

- Detect cyber incidents.
- Investigate affected systems.
- Isolate compromised devices.
- Restore systems and verify security.
- Monitor the network after recovery.

Incident Response Manager

- Lead the entire response process.
- Coordinate all departments.
- Approve containment and recovery actions.
- Maintain incident documentation.

Senior Management

- Approve major operational decisions.
- Allocate financial and technical resources.
- Communicate with external stakeholders.
- Support business continuity planning.

Legal and Compliance Team

- Ensure compliance with privacy regulations.
- Handle breach notification requirements.
- Preserve digital evidence for investigations.

Human Resources

- Communicate with employees.
- Coordinate awareness activities.
- Handle insider-related investigations if necessary.

Public Relations Team

- Prepare official public statements.
 - Communicate with students, parents, and media.
 - Protect the university's reputation.
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5. Benefits of an Escalation Matrix

- Ensures rapid reporting of incidents.
 - Clearly defines responsibilities.
 - Reduces communication delays.
 - Improves coordination between departments.
 - Supports legal and regulatory compliance.
 - Minimizes business and operational impact.
 - Enhances incident response efficiency and recovery.
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Conclusion

A well-designed Escalation Matrix and secure communication process help universities respond quickly to cyberattacks while protecting sensitive student information. Clearly defined reporting levels, responsibilities, and communication channels improve coordination, reduce confusion, ensure regulatory compliance, and support faster recovery from security incidents.